

M14 / ASSESSMENT

M14 Module Revision Test

Standalone Zero-to-Job-Ready Data Engineer 2026

Study, practise, check and explain.

A complete learning route

Start with the lesson video. Read the worked explanation beside it, pause to reproduce the example, then solve the supplied task. The books contain your study notes; keep your own work and results as evidence.

Use the protected answers after saving your first attempt. A changed retry shows what you can now do independently. There is no forced timer during learning.

One course. M00-M19. 264 lessons and 50 project phases.

Contents

Tap a topic or use the PDF bookmarks. These numbers are the actual pages in this edition.

Read this first 3

Read this first

No forced timer. Close learner book, videos, Quick Revision and answers. Save a genuine attempt before opening protected answers.

1. Batch vs streaming: A logistics company needs instant route alerts but only a nightly billing total. Design the boundary and justify why one mode should not replace the other.
2. Events/topics: Explain why two independent consumer applications can read the same retained Kafka topic without stealing records from each other.
3. Producers/keys: Choose a key for order lifecycle events. Explain ordering gained and one skew/partition-count risk.
4. Consumers: Give the safe processing/commit order for an external database sink and identify the replay window.
5. Partitions: A four-partition topic shows status B before status A in a merged dashboard. Under what condition is that legal even if Kafka preserved each partition order?
6. Offsets/replay: Explain why `auto.offset.reset=earliest` may do nothing for an existing healthy consumer group.
7. Consumer groups: What happens when 10 consumers join one group reading a 6-partition topic? What changes if they use 10 different group IDs?
8. Delivery: Distinguish producer idempotence, Kafka transactions and application/sink idempotency.
9. Schema evolution: Give one additive compatible change and one breaking change, and state the migration idea.
10. Python workflow: Why is `produce()` returning not delivery proof? What evidence should the script save?
11. Spark integration: Explain Kafka source offsets versus Spark checkpoint state and why deleting a checkpoint can be dangerous.
12. Operations: Only partition 2 has growing lag. Give a diagnostic order that does not begin with "restart everything".